



VITYARTHI PROJECT REPORT

SUBJECT: FUNDAMENTALS OF AIML

SMART WASTE MANAGEMENT



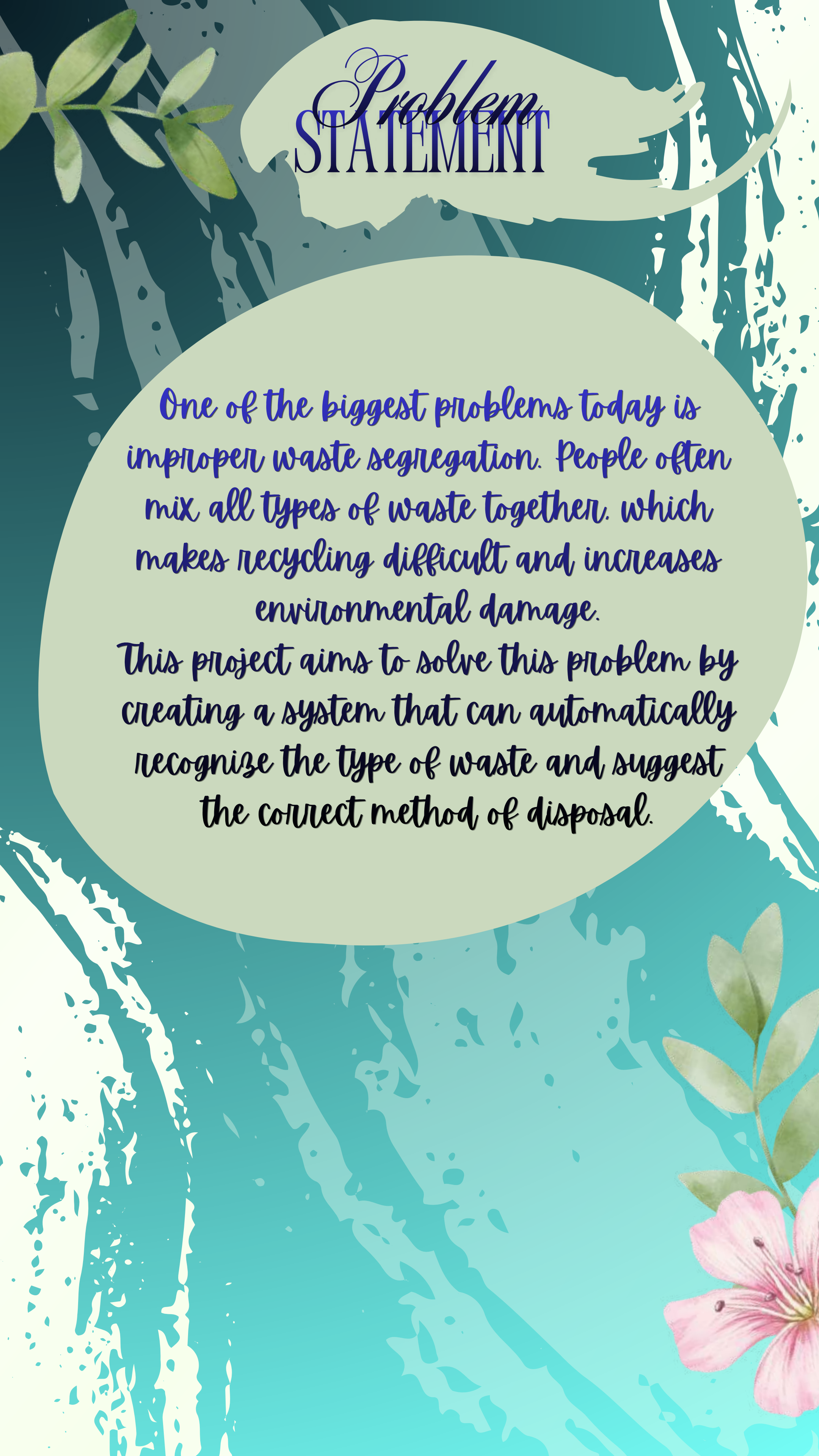
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INTRODUCTION

- Nowadays, waste management has become a serious issue in our daily lives. We often see garbage being thrown without proper segregation, which leads to pollution and health problems. Many people are not aware of how to dispose of waste correctly.
- With the help of Artificial Intelligence, we can make this process easier and more efficient. This project focuses on building a system that can identify different types of waste using images and guide users on how to dispose of them properly.

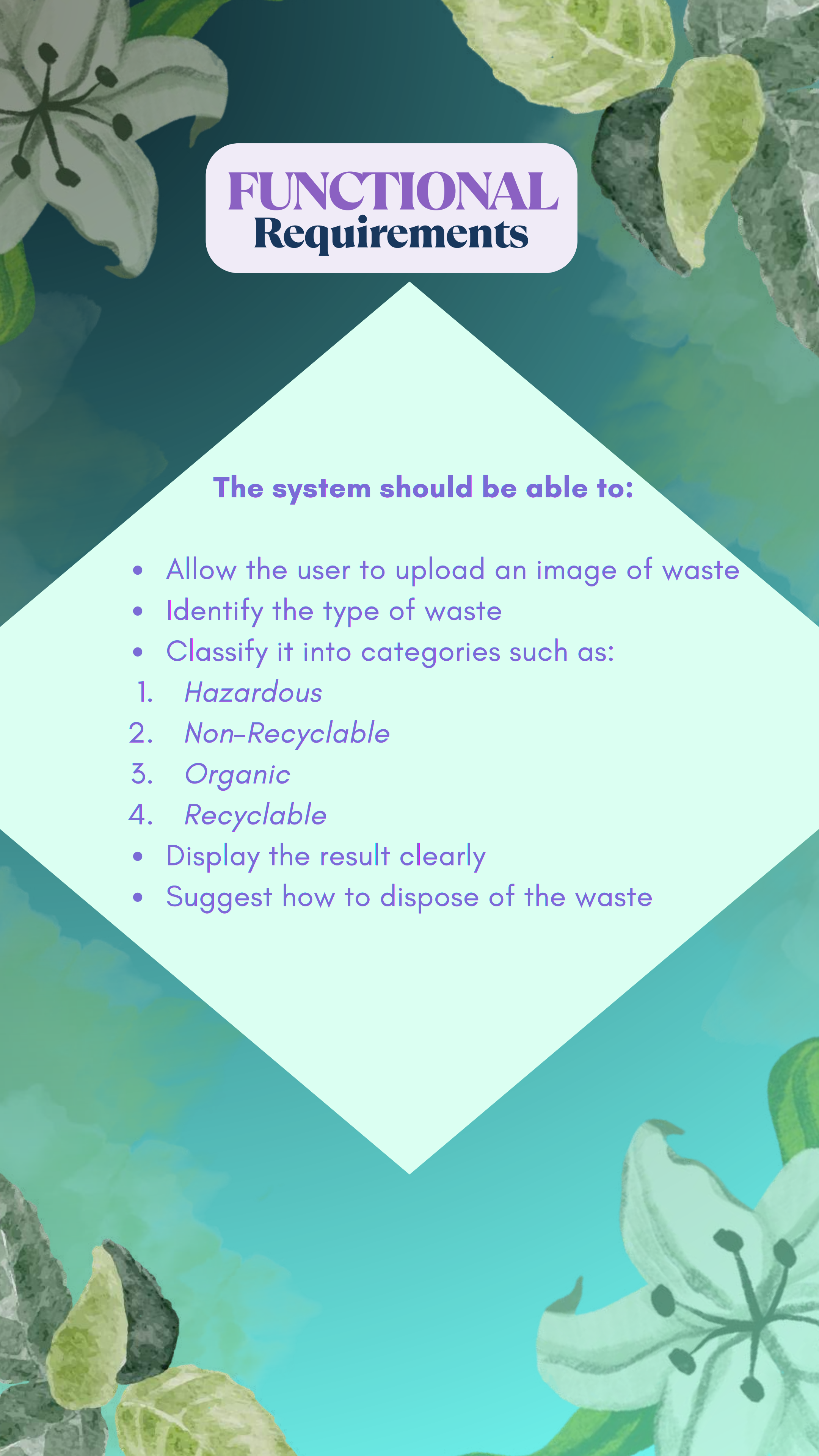




Problem STATEMENT

One of the biggest problems today is improper waste segregation. People often mix all types of waste together, which makes recycling difficult and increases environmental damage.

This project aims to solve this problem by creating a system that can automatically recognize the type of waste and suggest the correct method of disposal.



FUNCTIONAL Requirements

The system should be able to:

- Allow the user to upload an image of waste
- Identify the type of waste
- Classify it into categories such as:
 1. *Hazardous*
 2. *Non-Recyclable*
 3. *Organic*
 4. *Recyclable*
- Display the result clearly
- Suggest how to dispose of the waste

Non-Functional **REQUIREMENTS**

- The system should be easy to use
- It should give results quickly
- The predictions should be reasonably accurate
- The interface should be simple and user-friendly



System Architecture

The system is made up of three main parts:

1. User Interface – where users upload images
2. Machine Learning Model – which processes the image
3. Dataset – used to train the model

The overall process is simple:

User uploads an image → the system processes it → the model predicts → result is shown.

Design Diagrams

Workflow Diagram

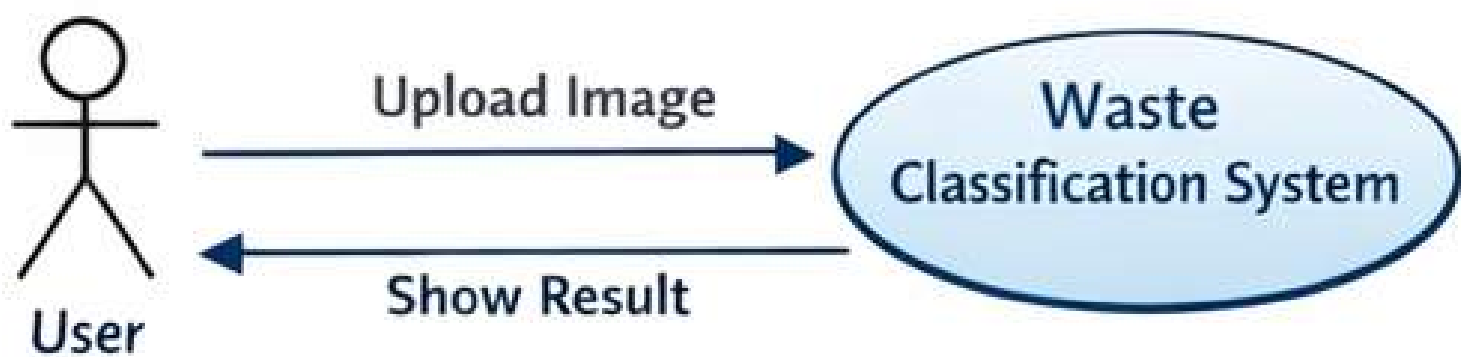
Workflow Diagram



This workflow represents the step-by-step process from image upload to final prediction output.

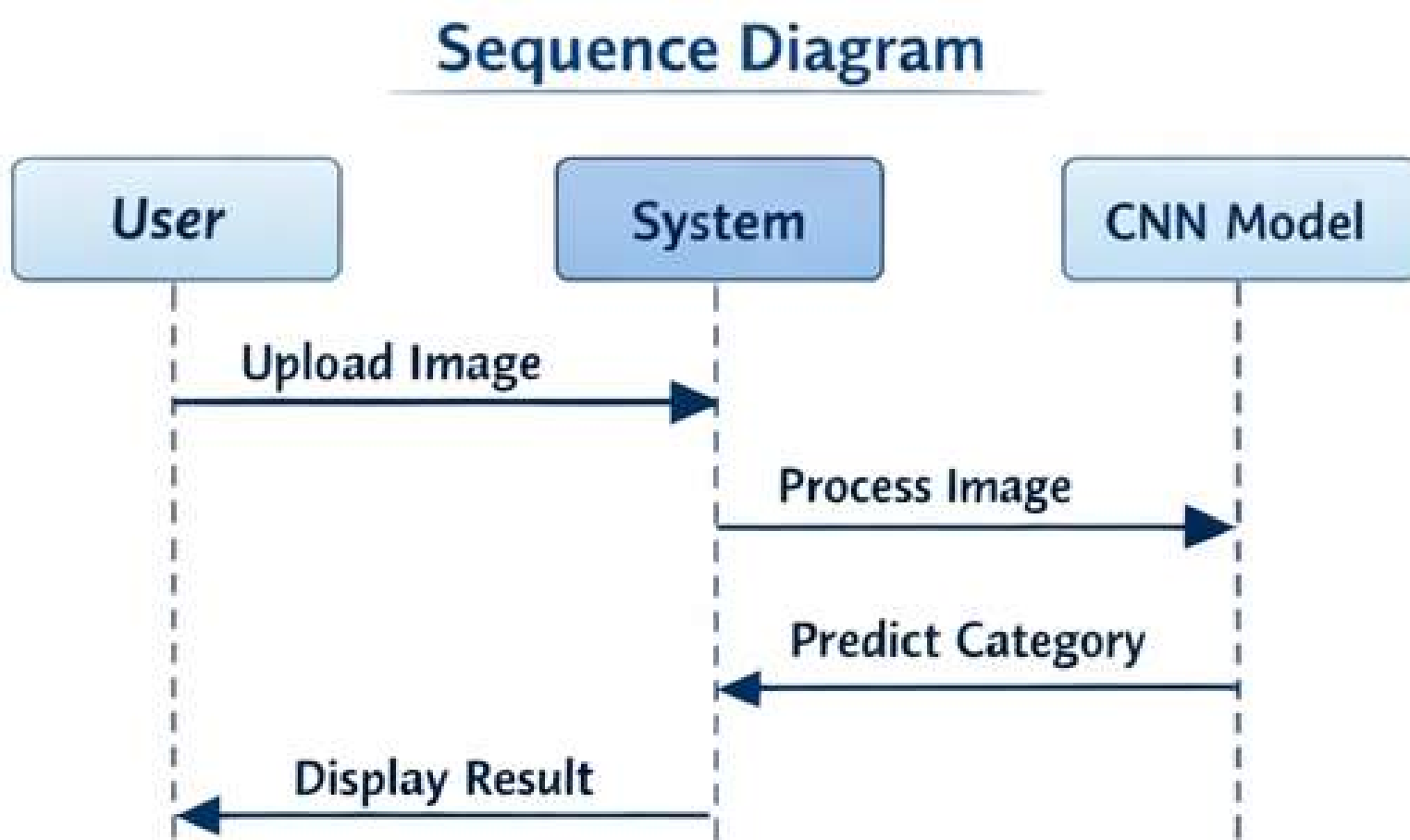
Use Case Diagram

Use Case Diagram



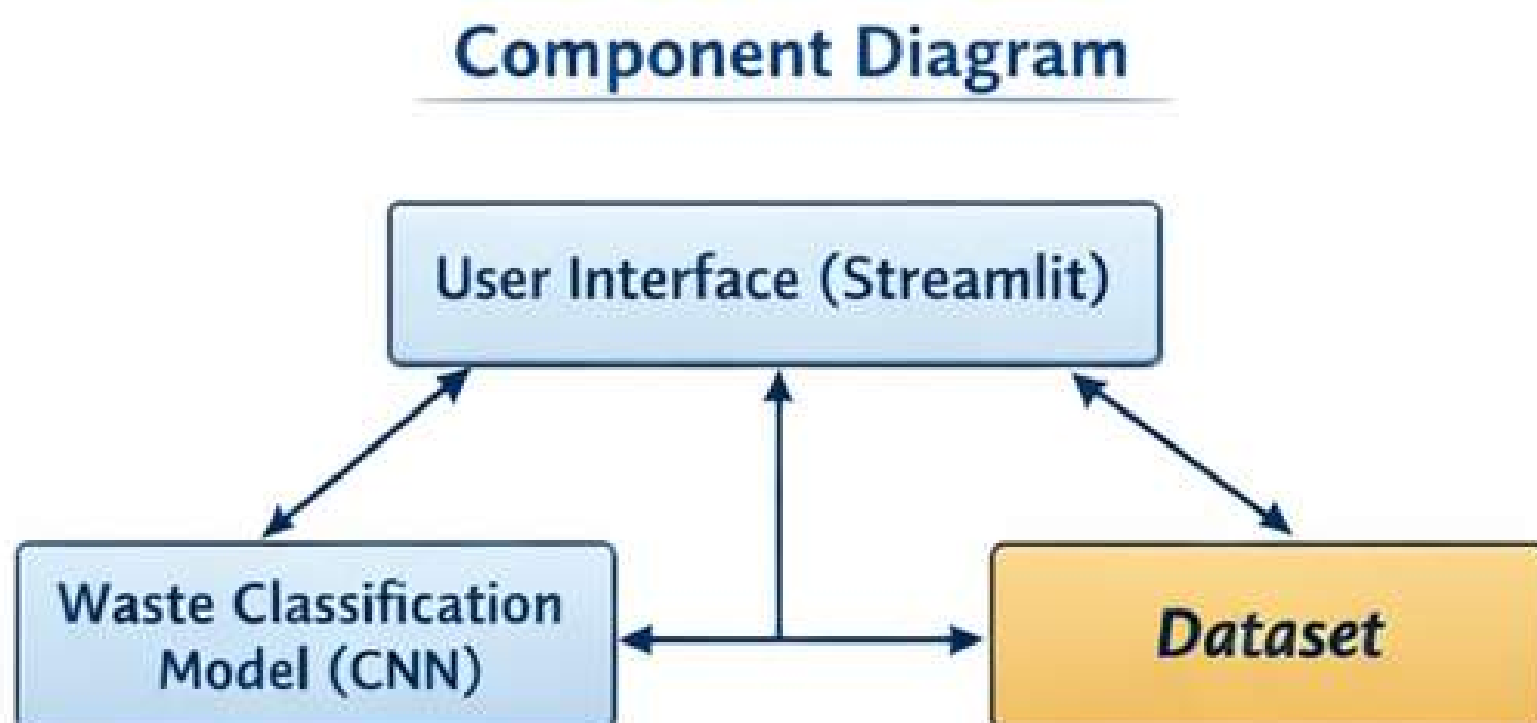
This diagram shows how the user interacts with the system by uploading an image and receiving waste classification results.

Sequence Diagram



This diagram explains how the user, system, and model interact sequentially during prediction

Component Diagram



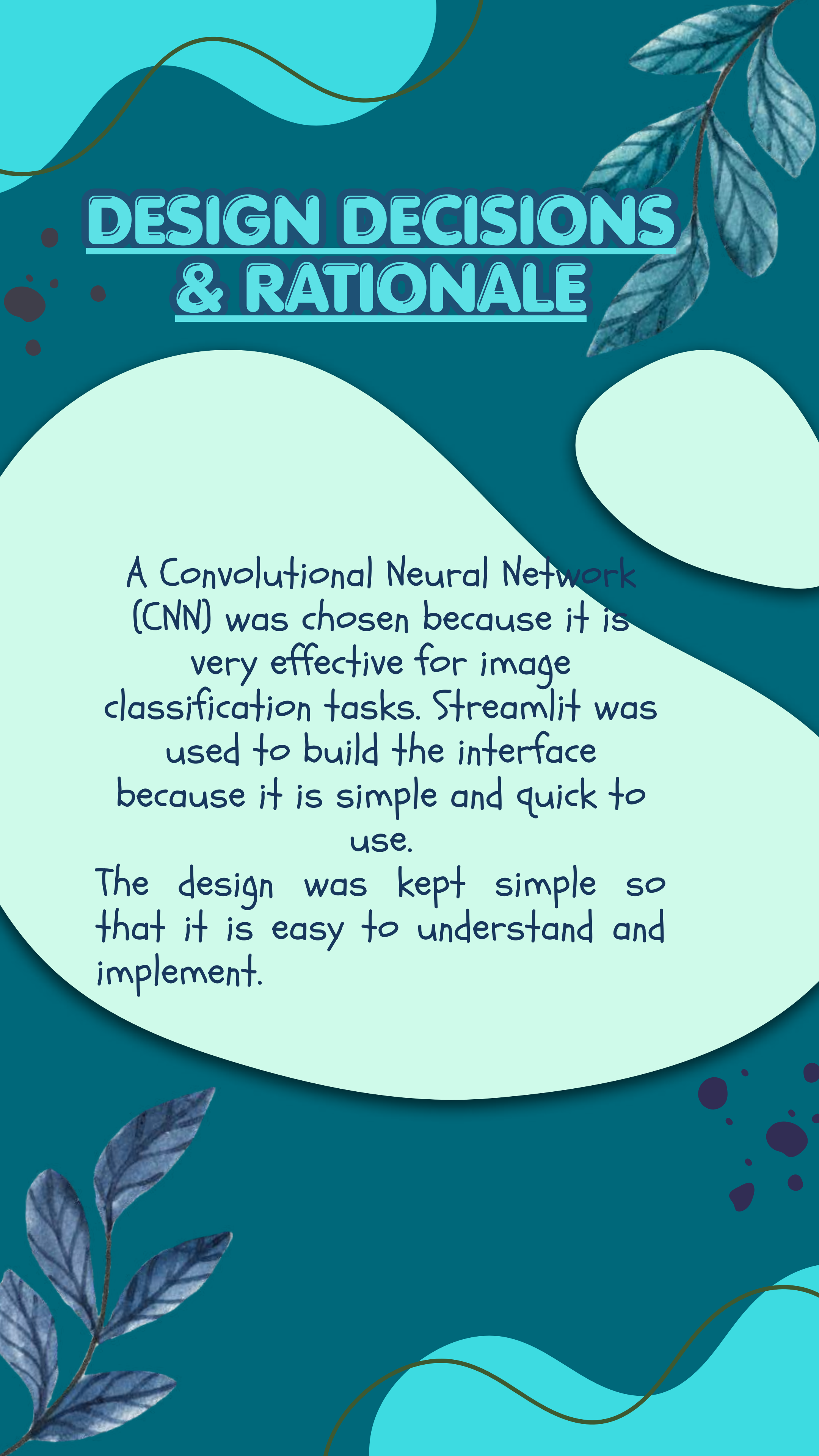
This diagram shows the main components of the system such as UI, model, and dataset.

ER Diagram

ER Diagram



This ER diagram represents the relationship between user, uploaded images, and prediction results.



DESIGN DECISIONS & RATIONALE

A Convolutional Neural Network (CNN) was chosen because it is very effective for image classification tasks. Streamlit was used to build the interface because it is simple and quick to use.

The design was kept simple so that it is easy to understand and implement.

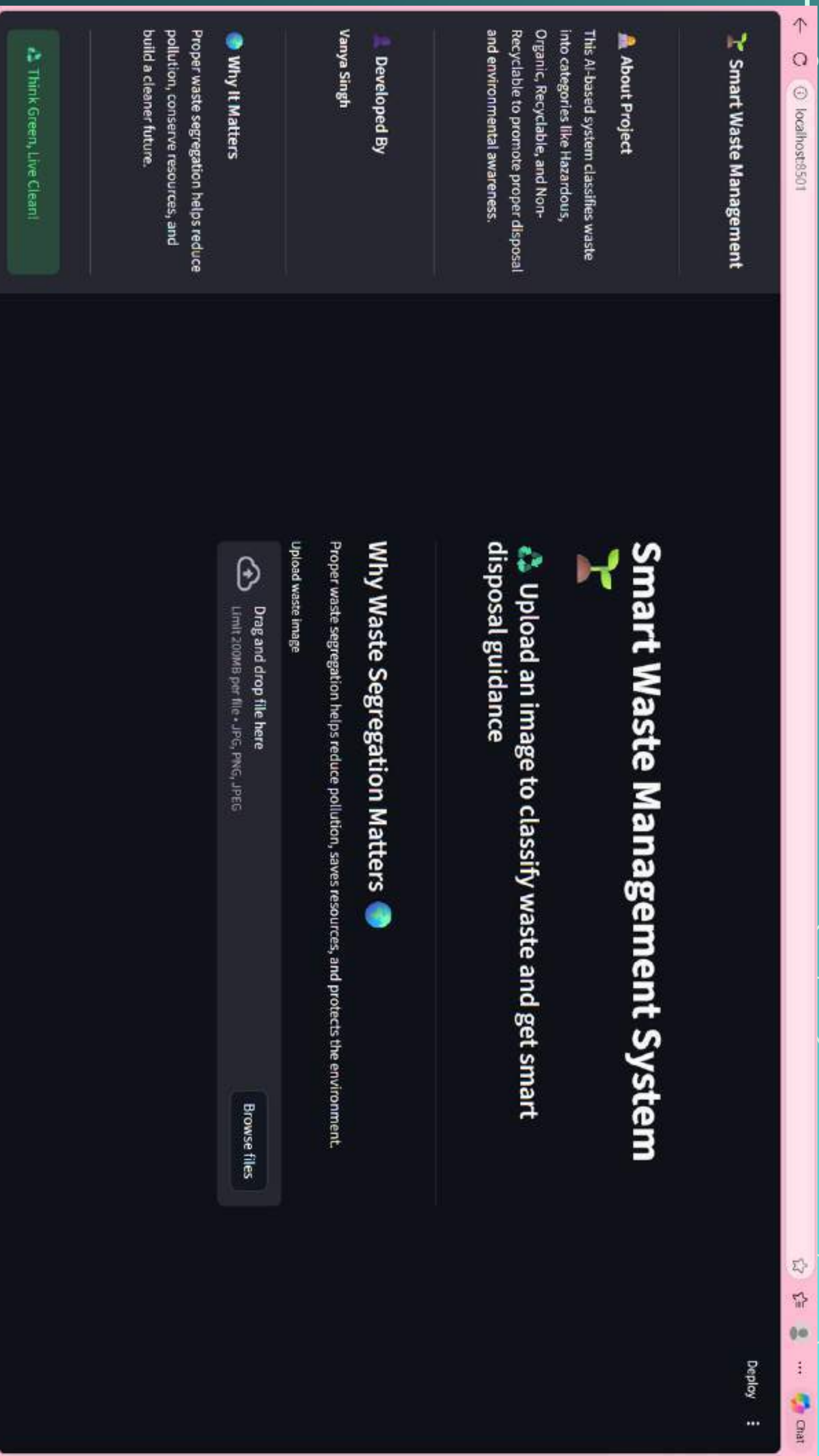
Implementation Details

The project is implemented using Python. Various libraries were used, such as TensorFlow for building the model, NumPy for calculations, Pillow for image processing, and Streamlit for the user interface.

The model takes an image as input and classifies it into one of four categories. The output is then shown to the user along with a suggestion.



Screenshot RESULTS



Why Waste Segregation Matters 🌍

Proper waste segregation helps reduce pollution, saves resources, and protects the environment.

Upload waste image



Drag and drop file here

Limit 200MB per file • JPG, PNG, JPEG

Browse files



Screenshot 2026-03-20 185545.png 342.3KB



Uploaded Image

Prediction: Organic

Dispose in compost/wet waste bin 🗑️

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 Drag and drop file here
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Browse files

 Screenshot 2026-03-20 195727.png 496.3KB

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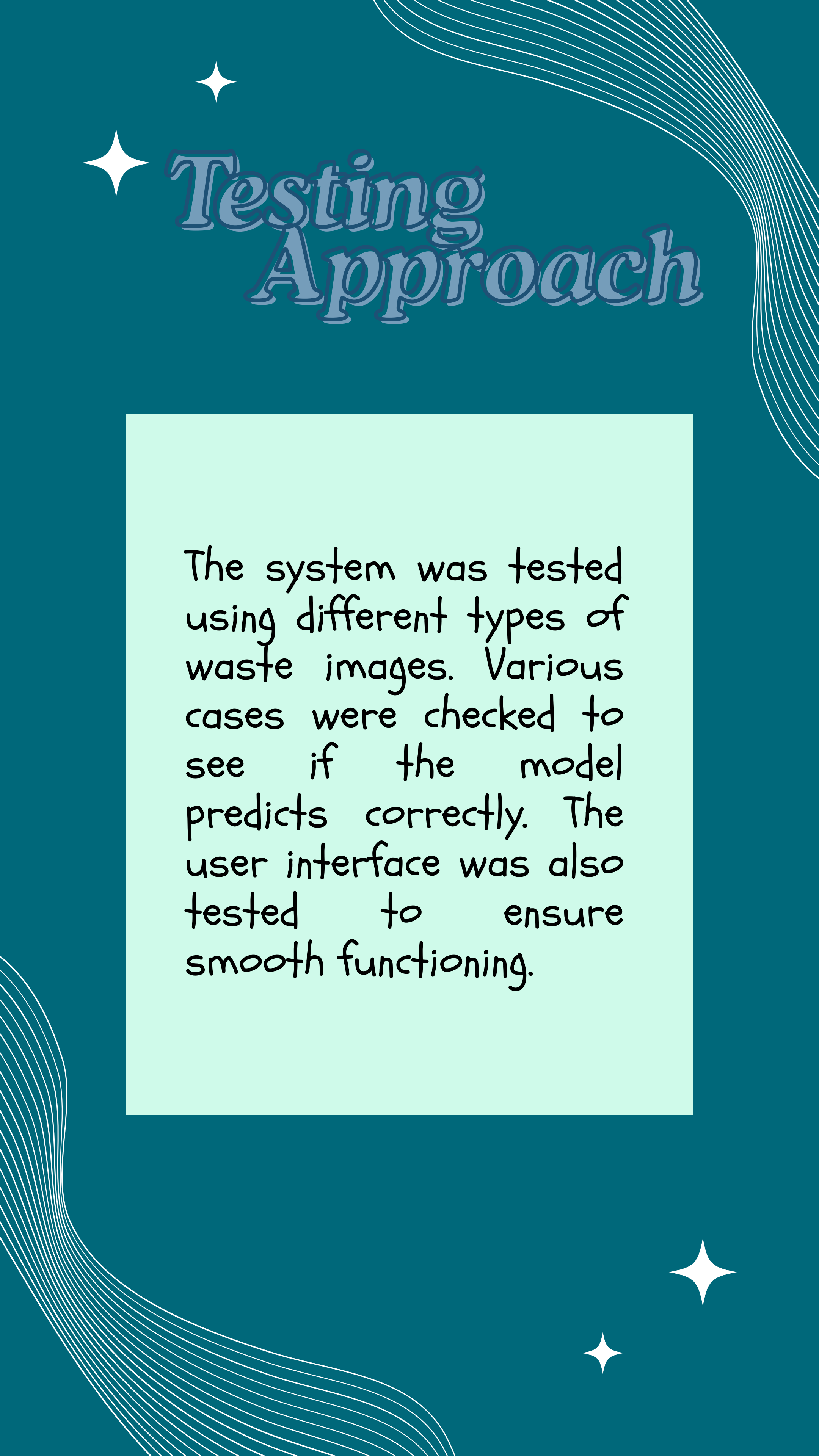
Uploaded Image

Prediction: Hazardous

Prediction: Hazardous

Handle carefully - hazardous waste 

“The system successfully classifies waste images and provides appropriate disposal suggestions. Different test cases were used to verify the accuracy of predictions.”



Testing Approach

The system was tested using different types of waste images. Various cases were checked to see if the model predicts correctly. The user interface was also tested to ensure smooth functioning.



Challenges Faced

- Limited number of images in the dataset
- Some waste items look very similar
- Improving the accuracy of the model

Learning & Key Takeaways

Through this project, I learned how Machine Learning can be used in real-life applications. I also understood how to work with datasets, train models, and build a simple user interface.

Future Enhancements

- Converting the system into a mobile application
 - Using live camera for real-time detection
 - Improving accuracy with more data
 - Adding more waste categories
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REFERENCES

- Kaggle Dataset
- TensorFlow Documentation
- Streamlit Documentation
- Python Documentation





Thank
You